Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A half-metre long tube open at one end, with a movable piston shows resonance with a tuning fork of frequency $512\ \text{Hz}$ when the tube length is $16.0\ \text{cm}$ and $49.0\ \text{cm}$. Calculate the speed of sound at the temperature of the experiment and determine the endcorrection.

(A) 338 ms^{-1}, 0.5 cm

(B) 383 ms^{-1}, 1.5 cm

(C) 335 ms^{-1}, 0.5 cm

(D) 332 ms^{-1}, 1.5 cm

Q.2

Standing waves are produced by the superposition of two waves

$$y_1 = 0.05\sin(3\pi t - 2x)$$
$$\text{and} \quad y_2 = 0.05\sin(3\pi t + 2x)$$

where x and y are expressed in metres and t is in seconds. What is the amplitude of a particle at $x = 0.5\ \text{m}$. Given $\cos(57.3^\circ) = 0.54$

(A) $2.7\ \text{cm}$

(B) $5.4\ \text{cm}$

(C) $8.1\ \text{cm}$

(D) $10.8\ \text{cm}$

Q.3

A sonometer wire AB of length $110\ \text{cm}$ is fixed at both ends. How far from ends A and B should the two bridges be placed below the wire so that the fundamental frequencies of the three segments of the wire are in the ratio $1:2:3$?

(A) $60\ \text{cm}, 20\ \text{cm}$

(B) $50\ \text{cm}, 30\ \text{cm}$

(C) $45\ \text{cm}, 35\ \text{cm}$

(D) $40\ \text{cm}, 30\ \text{cm}$

Q.4

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A wire of density $9\ \mathrm{g\ cm^{-3}}$ is stretched between two clamps $100\ \mathrm{cm}$ apart while being subjected to an extension of $0.05\ \mathrm{cm}$. What is the lowest frequency of transverse vibrations of the wire, assuming that the Young's modulus of the material of the wire = $0.9 \times 10^{11}\ \mathrm{N\ m^{-2}}$

(A) 38.3 Hz

(B) 35.3 Hz

(C) 37.3 Hz

(D) 36.3 Hz

Q.5

Standing waves are produced by the superposition of two waves
$y_1 = 5\sin(3\pi t - 2\pi x)$
and $y_2 = 5\sin(3\pi t + 2\pi x)$
where y and x are in cm and t in second. Find the amplitude of the particle at $x = 2\ \mathrm{cm}$.

(A) 10 cm

(B) 12 cm

(C) 14 cm

(D) 16 cm

Q.6

If $\mathrm{n_1}, \mathrm{n_2}$ and $\mathrm{n_3}$ are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequencyn of the string is given by

(A) $\frac{1}{\mathrm{n}} = \frac{1}{\mathrm{n_1}} + \frac{1}{\mathrm{n_2}} + \frac{1}{\mathrm{n_3}}$

(B) $\frac{1}{\sqrt{\mathrm{n}}} = \frac{1}{\sqrt{\mathrm{n_1}}} + \frac{1}{\sqrt{\mathrm{n_2}}} + \frac{1}{\sqrt{\mathrm{n_3}}}$

(C) $\sqrt{\mathrm{n}} = \sqrt{\mathrm{n_1}} + \sqrt{\mathrm{n_2}} + \sqrt{\mathrm{n_3}}$

(D) $\mathrm{n} = \mathrm{n_1} + \mathrm{n_2} + \mathrm{n_3}$

Q.7

In the sonometer experiment, a tuning fork of frequency $256\ \mathrm{Hz}$ is in resonance with $0.4\ \mathrm{m}$ length of the wire when the iron load attached to free end of wire is $2\ \mathrm{kg}$. If the load is immersed in water, the length of the wire in resonance would be (specific gravity of iron $= 8$)

(A) $0.37\ \mathrm{m}$

(B) $0.43\ \mathrm{m}$

(C) $0.31\ \mathrm{m}$

(D) $0.2\ \mathrm{m}$

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Q.8

The length of the wire between two ends of a sonometer is $100\ \text{cm}$. What should be the positions of two bridges below the wire so that the three segments of the wire have their fundamental frequencies in the ratio of $1:3:5$?

(A) $\frac{1500}{23}\ \text{cm}, \frac{2000}{23}\ \text{cm}$

(B) $\frac{1500}{23}\ \text{cm}, \frac{500}{23}\ \text{cm}$

(C) $\frac{1500}{23}\ \text{cm}, \frac{300}{23}\ \text{cm}$

(D) $\frac{300}{23}\ \text{cm}, \frac{1500}{23}\ \text{cm}$

Q.9

Length of a sonometer wire between two fixed ends is $110\ \text{cm}$. If the fundamental frequencies are in the ratio of $1:2:3$, then what is the ratio of lengths of these segments of the wire?

(A) $3:2:1$

(B) $6:3:2$

(C) $6:2:3$

(D) $2:3:6$

Q.10

A sonometer wire of length $1.5\ \text{m}$ is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are $7.7 \times 10^3\ \text{kg/m}^3$ and $2.2 \times 10^{11}\ \text{N/m}^2$ respectively?

(A) $188.5\ \text{Hz}$

(B) $178.2\ \text{Hz}$

(C) $200.5\ \text{Hz}$

(D) $770\ \text{Hz}$

Q.11

A sonometer wire supports a $4\ \text{kg}$ load anc vibrates in fundamental mode with a tuning fork of frequency $416\ \text{Hz}$. The length of the wire between the bridges is now doubled. In order to maintain fundamental mode, the load should be changed to

(A) $1\ \text{kg}$

(B) $2\ \text{kg}$

(C) $4\ \text{kg}$

(D) $16\ \text{kg}$

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Q.12

A string is stretched between fixed points separated by $75.0\ \text{cm}$. It is observed to have resonant frequencies of $420\ \text{Hz}$ and $315\ \text{Hz}$. There are noother resonant frequencies between these two. Then, the lowest resonant frequency for this string is

(A) $105\ \text{Hz}$

(B) $1.05\ \text{Hz}$

(C) $1050\ \text{Hz}$

(D) $10.5\ \text{Hz}$

Q.13

The frequency of a tuning fork is 384sec^{-1} and velocity of sound in air is $352\ \text{m/s}$. Find how far the sound has traversed while fork completes 36 vibrations?

(A) $3\ \text{m}$

(B) $13\ \text{m}$

(C) $23\ \text{m}$

(D) $33\ \text{m}$

Q.14

The fundamental frequency of a sonometer wire of length ℓ is f_0. A bridge is now introduced at a distance of $\Delta\ell$ from the centre of the wire $(\Delta\ell << \ell)$. The number of beats heard if both sides of the bridges are set into vibration in their fundamental modes are -

(A) $\frac{8f_0\Delta\ell}{\ell}$

(B) $\frac{f_0\Delta\ell}{\ell}$

(C) $\frac{2f_0\Delta\ell}{\ell}$

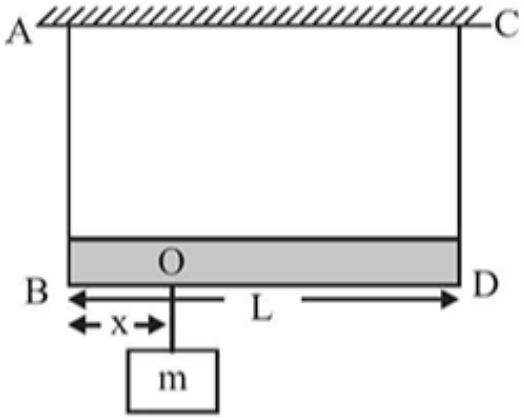
(D) $\frac{4f_0\Delta\ell}{\ell}$

Q.15

A massless rod of length L is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to ' x '. Further it is observed that the frequency of

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1 st harmonic in AB is equal to 2 nd harmonic frequency in $CD.'x$ ' is



(A) $\frac{L}{5}$

(B) $\frac{4L}{5}$

(C) $\frac{3L}{4}$

(D) $\frac{L}{4}$

Q.16

Two waves are represented by the equations $y_1 = a\sin(\omega t + kx + 0.57)m$ and $y_2 = a\cos(\omega t + kx)m$
where x is in meter and t in sec. The phase difference between them is

(A) 1.0 radian

(B) 1.25 radian

(C) 1.57 radian

(D) 0.57 radian

Q.17

The vibrations of a string of length $60\ \mathrm{cm}$ fixed at both the ends are represented by the equation $\mathrm{y} = 2\sin\left(\frac{4\pi \mathrm{x}}{15}\right)\cos(96\pi \mathrm{t})$ where x and y are in cm. The maximum number of loops that can be formed in it is

(A) 4

(B) 16

(C) 5

(D) 15

Q.18

In a resonance pipe the first and second resonance are obtained at depths 22.7 cm and 70.2 cm respectively. What will be the end correction?

(A) $1.05\ \mathrm{cm}$

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(B) 115.5 cm

(C) 92.5 cm

(D) 113.5 cm

Q.19

A source of the sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 m s^{-1} and in the air, it is 300 m s^{-1} The frequency of sound recorded by an observer who is standing in the air is

(A) 200 Hz

(B) 3000 Hz

(C) 120 Hz

(D) 600 Hz

Q.20

An observer moves towards a stationary source of sound with a speed $\frac{1}{5}^{th}$ of the speed of sound. The wavelength and frequency of the source emitted are λ and f respectively. The apparent frequency and wavelength recorded by the observer are respectively

(A) $1.2f,\ 1.2\lambda$

(B) $1.2f,\ \lambda$

(C) $f,\ 1.2\lambda$

(D) 0.8 f, 0.8λ

Answers & Solutions

Q.1 Answer:

338 ms^{-1}, 0.5 cm

Solution:

The tuning fork is in resonance at two lengths of the pipe, viz. 16 cm **and** 49 cm**. Since the second resonance length is about three times the first, it is clear that the fork is in resonance with the first harmonic (i.e. the fundamental mode) when** $L_1 = 16$ cm **and with the third harmonic when** $L_2 = 49$ cm**. Thus we have**

$N = v_1 = \frac{v}{4(L_1+e)}$ **(i)**

Also $N = v_3 = \frac{3v}{4(L_2+e)}$ **(ii)**

where N **is the frequency of tuning fork and** e **is the endcorrection. Dividing (ii) by (i) we get**

$3(16.0 + e) = 49.0 + e$ or $e = 0.5$ cm

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Then the speed of sound as obtained from (i) is

$$\begin{aligned} v &= 4N\,(L_1 + e) \\ &= 4 \times 512 \times (16.0 + 0.5) \\ &= 33792 \text{ cm s}^{-1} = 337.92 \text{ m s}^{-1} \\ &\simeq 338 \text{ m s}^{-1} \end{aligned}$$

Q.2 Answer:

5.4 cm

Solution:

The resultant displacement is given by $y = y_1 + y_2 = 0.05\{\sin(3\pi t - 2x) + \sin(3\pi t + 2x)\}$

Using the trigonometric relation $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2\sin\alpha\cos\beta$

we have $y = 0.1\cos 2x \cdot \sin 3\pi t$

or $y = R\sin 3\pi t$

where R**, the amplitude of standing waves, is given by**

$$R = 0.1\cos 2x$$

$$\text{When } x = 0.5 \text{ m} \quad \cos 2x = \cos(2 \times 0.5\text{rad})$$

$$= \cos(1\text{rad})$$

$$= \cos 57.3^\circ = 0.54$$

$\therefore$ **Amplitude** R **at** $x = 0.5$ **is** $0.1 \times 0.54 = 0.054 \text{ m} = 5.4 \text{ cm}$

Q.3 Answer:

60 cm, 20 cm

Solution:

Let L_1, L_2 **and** L_3 **be the lengths of the three segments of wire. Their fundamental frequencies are**

$$v_1 = \frac{1}{2L_1}\sqrt{\frac{T}{m}}$$

$$v_2 = \frac{1}{2L_2}\sqrt{\frac{T}{m}}$$

$$v_3 = \frac{1}{2L_2}\sqrt{\frac{T}{m}}$$

which give $v_1 L_1 = v_2 L_2 = v_3 L_3$**. Since** $v_1 : v_2 : v_3 = 1 : 2 : 3$**, we have** $v_2 = 2v_1$ **and** $v_3 = 3v_1$**. Hence**

$$v_1 L_1 = 2v_1 L_2 \text{ and } \quad v_1 L_1 = 3v_1 L_3 \text{ so that}$$

$$L_2 = L_1\frac{v_1}{2v_1} = \frac{L_1}{2}$$

$$\text{and} \quad L_3 = \frac{L_1}{3}$$

$$\text{But} \quad L_1 + L_2 + L_3 = L$$

$$\Rightarrow \quad L_1\left(1 + \frac{1}{2} + \frac{1}{3}\right) = 110$$

$$\Rightarrow \quad L_1 = 60 \text{ cm}$$

Hence $L_3 = \frac{60}{3} = 20 \text{ cm}$

Hence the correct choice is (a).

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Q.4 Answer:

35.3 Hz

Solution:

Young's modulus

$Y = \frac{\text{stress}}{\text{strain}} = \frac{T/A}{l/L} = \frac{TL}{Al}$

where $T =$ **tension,** $L =$ **original length** $= 100$ cm, $l =$ **extension** $= 0.05$ cm **and** $A =$ **area of cross-section of the wire.**

Hence, $\frac{T}{A} = \frac{Yl}{L} = \frac{0.9\times10^{11}\times0.05}{100} = 4.5 \times 10^7$

Now mass per unit length $m = \frac{\text{mass}}{\text{length}}$

$= \frac{\text{volume} \times \text{density}}{\text{length}}$

$= \frac{\text{area} \times \text{length} \times \text{density}}{\text{length}}$

$= \text{area} \times \text{density}$

$= A\rho$

Density $\rho = 9$ g cm$^{-3} = 9000$ kg m^{-3}**. The lowest frequency is the frequency of the fundamental mode.**

$$v_1 = \frac{1}{2L}\sqrt{\frac{T}{m}} = \frac{1}{2L}\sqrt{\frac{T}{A\rho}}$$

$$= \frac{1}{2 \times 1} \times \sqrt{\frac{4.5 \times 10^7}{9000}}$$

$$= 35.3 \text{ Hz}$$

Q.5 Answer:

10 cm

Solution:

$y = y_1 + y_2$

$= 5[\sin(3\pi t - 2\pi x) + \sin(3\pi t + 2\pi x)]$

Using $\sin(\alpha+\beta) + \sin(\alpha-\beta) = 2\sin\alpha\cos\beta$**, we get**

$$y = 10\cos(2\pi x)\sin(3\pi t)$$

$\Rightarrow y = A\sin(3\pi t),$ where $A = 10\cos(2\pi x)$

Amplitude A **at** $x = 2$ cm **is**

$10 \times \cos(2\pi \times 2) = 10\cos 4\pi = 10$ cm

Q.6 Answer:

$\frac{1}{\text{n}} = \frac{1}{\text{n}_1} + \frac{1}{\text{n}_2} + \frac{1}{\text{n}_3}$

Solution:

Total length of string $\ell = \ell_1 + \ell_2 + \ell_3$ **(As string is divided into three segments) But frequency** $\propto \frac{1}{\text{length}}$ $\left(\because \text{f} = \frac{1}{2\ell}\sqrt{\frac{\text{T}}{\text{m}}}\right)$

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so $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

Q.7 Answer:

$0.37\ \mathrm{m}$

Solution:

The tension in the string initially $= mg = Vdg$

where, $m = 2\ \mathrm{kg}$

Tension after immersion in water

$= mg - V\rho g = Vdg - V\rho g = Vd\left(1 - \frac{\rho}{d}\right) g$

$= mg\left(1 - \frac{\rho}{8}\right) = \frac{7}{8} mg$

Since the frequency remains the same,

$\frac{\sqrt{T_1}}{l_1} = \frac{\sqrt{T_2}}{l_2}$

$\Longrightarrow \frac{\sqrt{mg}}{0.4} = \frac{\sqrt{\frac{7}{8} mg}}{l_2}$

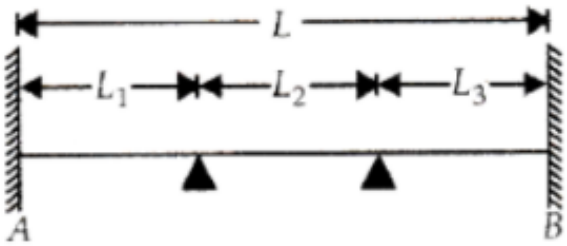
$\Longrightarrow l_2 = 0.37\ \mathrm{m}$

Q.8 Answer:

$\frac{1500}{23}\ \mathrm{cm}, \frac{2000}{23}\ \mathrm{cm}$

Solution:

Let $L\ (= 100\ \mathrm{cm}\ )$ **be the length of the wire and** L_1, L_2 **and** L_3 **are the lengths of the segments as shown in the figure.**



Fundamental frequency, $v \propto 1/L$

As the fundamental frequencies are in the ratio of $1 : 3 : 5$**,**

$$\therefore \quad L_1 : L_2 : L_3 = \frac{1}{1} : \frac{1}{3} : \frac{1}{5} = 15 : 5 : 3$$

Let x **be the common factor. Then**

$$15x + 5x + 3x = L = 100$$
$$23x = 100 \text{ or } x = \frac{100}{23}$$
$$\therefore \quad L_1 = 15 \times \frac{100}{23} = \frac{1500}{23}\ \mathrm{cm}$$
$$L_2 = 5 \times \frac{100}{23} = \frac{500}{23}\ \mathrm{cm};$$
$$L_3 = 3 \times \frac{100}{23} = \frac{300}{23}\ \mathrm{cm}$$

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$\therefore$ The bridges should be placed from A at $\frac{1500}{23}$ cm and $\frac{2000}{23}$ cm respectively.

Q.9 Answer:

$6 : 3 : 2$

Solution:

$\mathrm{f_1 : f_2 : f_3 = 1 : 2 : 3}$

$\Rightarrow \frac{1}{\mathrm{l_1}} : \frac{1}{\mathrm{l_2}} : \frac{1}{\mathrm{l_3}} = 1 : 2 : 3$

$\Rightarrow \mathrm{l_1 : l_2 : l_3 = 6 : 3 : 2}$

$\mathrm{l_1} = \left(\frac{6}{2+3+6}\right) \times 110 = 60 \text{ cm}$

$\mathrm{l_2} = \frac{3}{11} \times 110 = 30 \text{ cm}$

$\mathrm{l_3} = \frac{2}{11} \times 110 = 20 \text{ cm}$

Q.10 Answer:

178.2 Hz

Solution:

Fundamental frequency $\mathrm{f} = \frac{1}{2\mathrm{l}}\sqrt{\frac{\mathrm{T}}{\mu}}$

$= \frac{1}{2\mathrm{l}}\sqrt{\frac{\mathrm{T}}{\mathrm{Ap}}}$

$\frac{1}{2\mathrm{l}}\sqrt{\frac{\text{stress}}{\mathrm{p}}} = \frac{1}{2\times 1.5}\sqrt{\frac{2.2\times 10^{11}\times 10^{-2}}{7.7\times 10^{3}}} = 178.2 \text{ Hz}$

Q.11 Answer:

16 kg

Solution:

We know that

fundamental trequency fo $= \frac{v}{4l}$

$$f_0 = \frac{1}{4\ell}\sqrt{\frac{T}{\mu}}$$

According to Sonometer

$$T = mg$$

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After increasing length $l' = 2l$

$$f = \frac{1}{4\ell'}\sqrt{\frac{T^1}{\mu}}$$
$$= \frac{1}{4,2\ell}\sqrt{\frac{T^1}{\mu}}$$

from Question

$$f_0 = f$$
$$\frac{1}{4/l}\sqrt{\frac{T}{\mu}} = \frac{1}{4\times 2l}\sqrt{\frac{T'}{\mu}}$$
$$\sqrt{\frac{mg}{\mu}} = \frac{1}{2}\sqrt{\frac{m'g}{\mu}} \quad (\therefore T = mg)$$
$$\sqrt{m} = \frac{1}{2}\sqrt{m'}$$

Squaring both Side $m' = 4 \times m$

$$m' = 4 \times m$$
$$= 4 \times u = 16 \text{ kg} \quad (m = 4kg)$$

Q.12 Answer:

105 Hz

Solution:

Let 315 Hz be the nth harmonic of the string.

Thus $\frac{\text{nc}}{2\text{l}} = 315$ Hz

and $\frac{(\text{n}+1)\text{c}}{2\text{l}} = 420$ Hz

$\implies \frac{\text{n}+1}{\text{n}} = \frac{4}{3}$

$\implies \text{n} = 3$

Thus lowest resonant frequency $= \frac{\text{c}}{2\text{l}} = \frac{315}{\text{n}}\text{Hz} = 105$ Hz

Q.13 Answer:

33 m

Solution:

$$\text{x} = \text{v.t} = \text{v} \times \frac{\text{no of complete vibration}}{\text{frequency offork}} = 325\text{x}\frac{36}{384} = 33 \text{ m}$$

Q.14 Answer:

$\frac{8\text{f}_0\Delta\ell}{\ell}$

Solution:

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$f_0 = \frac{v}{2\ell}$

Now beat frequency $= f_1 - f_2$

$= \frac{v}{2\left(\frac{\ell}{2}-\Delta\ell\right)} - \frac{v}{2\left(\frac{\ell}{2}+\Delta\ell\right)} = \frac{v}{2}\left[\frac{1}{\frac{\ell}{2}-\Delta\ell} - \frac{1}{\frac{\ell}{2}+\Delta\ell}\right]$

$= (f_0\ell)\left[\frac{2}{\ell-2\Delta\ell} - \frac{2}{\ell+2\Delta\ell}\right]$

$= 2f_0\ell\left[\frac{\ell+2\Delta\ell-\ell+2\Delta\ell}{\ell^2-4(\Delta\ell)^2}\right] \approx 2f_0\ell\left(\frac{4\Delta\ell}{\ell^2}\right) \approx \frac{8f_0\Delta\ell}{\ell}$

Q.15 Answer:

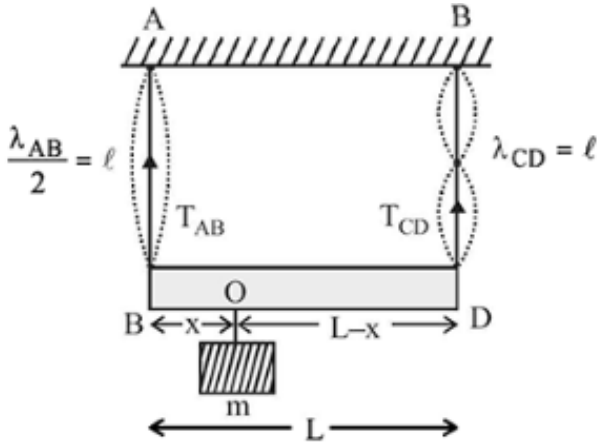
$\frac{L}{5}$

Solution:

Frequency of Ist harmonic of $AB = \frac{1}{2\ell}\sqrt{\frac{T_{AB}}{m}}$ **Frequency of 2 nd harmonic of** $CD = \frac{1}{\ell}\sqrt{\frac{T_{CD}}{m}}$

Given that the two frequencies are equal. $\therefore \quad \frac{1}{2\ell}\sqrt{\frac{T_{AB}}{m}} = \frac{1}{\ell}\sqrt{\frac{T_{CD}}{m}} \Rightarrow \frac{T_{AB}}{4} = T_{CD}$

(i) $\Rightarrow T_{AB} = 4T_{CD}$



For rotational equilibrium of massless rod, taking torque about point O.

(ii) $T_{AB} \times x = T_{CD}(L - x)$

For translational equilibrium, $T_{AB} + T_{CD} = mg$

(iii) On solving, (i) and (iii), we get $T_{CD} = \frac{mg}{5} \quad \therefore T_{AB} = \frac{4mg}{5}$

Substituting these values in (ii), we get $\frac{4mg}{5} \times x = \frac{mg}{5}(L - x) \Rightarrow x = \frac{L}{5}$

Q.16 Answer:

1.0 radian

Solution:

Here, $y_1 = a\sin(\omega t + kx + 0.57)$

and $y_2 = a\cos(\omega t + kx)$

$= a\sin\left[\frac{\pi}{2} + (\omega t + kx)\right]$

Phase difference, $\Delta\phi = \phi_2 - \phi_1$

$= \frac{\pi}{2} - 0.57$

$= \frac{3.14}{2} - 0.57 = 1.57 - 0.57 = 1$ **radian**

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Q.17 Answer:

16

Solution:

Let the string vibrates in p loops, wavelength of the pth mode of vibration is given by $\lambda_{\mathrm{p}} = \frac{2l}{\mathrm{p}}$

Given, $\mathrm{y} = 2\sin\left(\frac{4\pi \mathrm{x}}{15}\right)\cos(96\pi \mathrm{t})$

or $y = \left[\sin\left(\frac{4\pi x}{15} + 96\pi t\right) + \sin\left(\frac{4\pi x}{15} - 96\pi t\right)\right]$

Comparing it with standard equation, we get $v = \frac{96\pi}{2\pi} = 48\ \mathrm{Hz}$ **and** $\mathrm{k} = \frac{4\pi}{15}$

$\frac{1}{48} = \frac{2\times 60}{\mathrm{p}} \times \frac{4\pi}{15\times 96\pi}$

$\Rightarrow \mathrm{p} = 16$

Q.18 Answer:

1.05 cm

Solution:

For the end correction x,

$\Longrightarrow \quad x = \frac{l_2 - 3l_1}{2}$

$= \frac{70.2 - 3\times 22.7}{2} = 1.05\ \mathrm{cm}$

Q.19 Answer:

600 Hz

Solution:

The frequency is a characteristic of source. It is independent of the medium. Hence, the correct option is 600 Hz.

Q.20 Answer:

1.2f, λ

Solution:

Apparent frequency, $f' = \frac{v + v_0}{v} f$

$\Rightarrow f' = \frac{v + 1/5v}{v} f = 1.2f$

Wavelength does not change by motion of observer.

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